Unsloth

- Unsloth는 거대 언어 모델(LLM)의 파인튜닝(Fine-tuning) 속도를 획기적으로 높이고 메모리 사용량을 줄여주는 오픈소스 라이브러리입니다.
- 특히 개인용 GPU를 사용하는 학생이나 연구자들에게 매우 유용한 도구입니다.

Unsloth 장점

일반적으로 LLM을 학습시키려면 엄청난 양의 VRAM(그래픽 메모리)과 시간이 필요합니다. Unsloth는 이를 수학적으로 최적화하여 다음과 같은 이점을 제공합니다.

- `압도적인 속도` : 기존 Hugging Face 방식보다 2배~5배 더 빠른 학습 속도.
- `메모리 효율성` : 메모리 사용량을 70% 이상 절감. (고성능 GPU가 없어도 학습 가능!)
- `손실 없는 정확도` : 속도는 빨라지지만, 모델의 예측 정확도는 전혀 떨어지지 않습니다 (0% Accuracy Loss).
- `초보자 친화적` : 코랩(Google Colab)에서도 단 몇 줄의 코드로 실행 가능합니다.

성능 비교 (Llama-3 8B 모델 기준)

비교 항목	기본 방식 (Hugging Face)	Unsloth 적용 시
학습 시간	10시간	**약 2~3시간**
메모리(VRAM) 점유	16GB 이상	**약 7GB 미만**
정확도	100%	**100% (차이 없음)**

Unsloth LoRA / QLoRA 장점

관점	장점	설명
속도	**학습 속도 매우 빠름**	커스텀 CUDA 커널 + 최적화된 forward/backward
메모리	**VRAM 사용량 최소**	4bit 로딩 + QLoRA 기본 설계
난이도	**설정이 단순함**	YAML 없음, Python 몇 줄로 끝
실험	**빠른 반복 실험 가능**	하이퍼파라미터 변경 부담 낮음
하드웨어	**단일 GPU로 충분**	24GB GPU에서도 7B~13B 가능
통합	**Transformers 호환**	HF Trainer / PEFT 생태계 그대로 사용
저장	**HF Hub 업로드 간단**	`push_to_hub()` 바로 가능
디버깅	**에러 지점 명확**	분산/ZeRO 개념 없음

Unsloth - Fine Tuning

Dockerfile에 포함된 필수 요소

> RunPod에서 모델 학습 → Hugging Face Hub 저장

기능	라이브러리	상태
모델 로딩/학습	`unsloth`	✅
HF 모델 구조	`transformers`	✅
QLoRA	`bitsandbytes`	✅
Trainer	`trl`, `peft`	✅
Hub 업로드	`transformers` 내부	✅

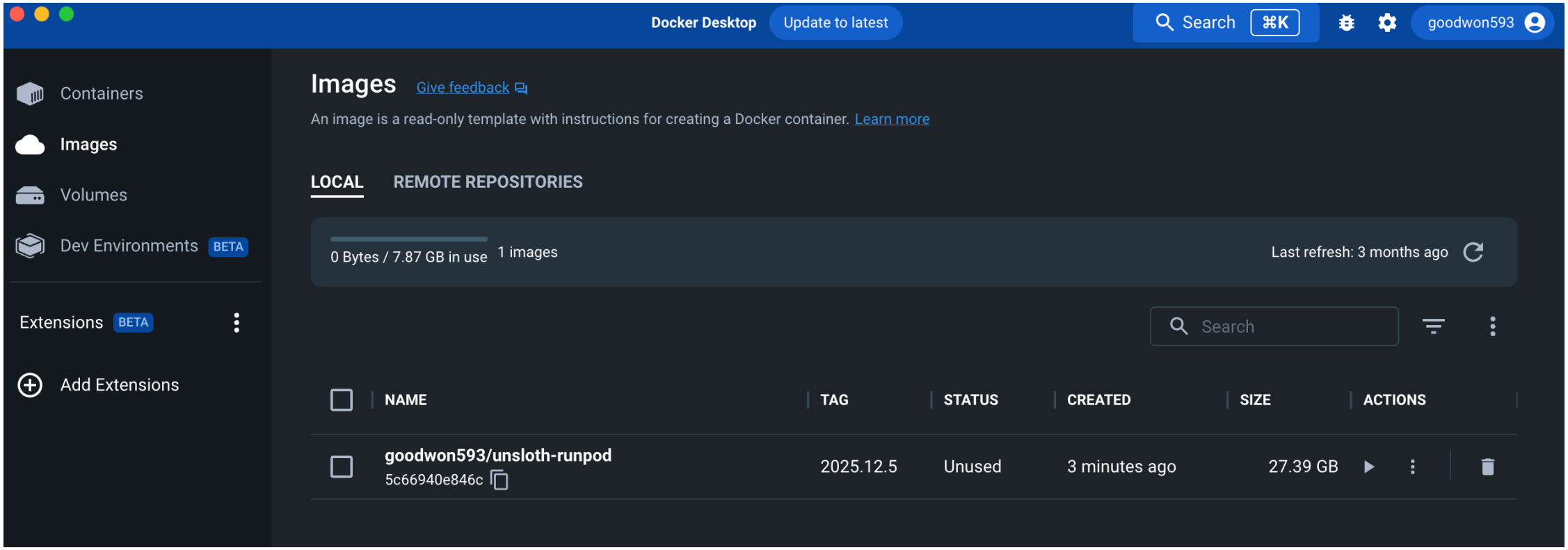
Docker Hub에 배포

단계1: Docker Image 생성

```
docker build -t <Docker ID>/unsloth-runpod:2025.12.5 .
```

```
Problems   Output   Debug Console   Terminal   Ports   Postman Console                    zsh

good593 🚀   ~/dev/github/course_LLM/2. LLM Fine Tuning/2. Fine Tuning with Runpod/3. 모델 학습/1. Unsloth - Fine Tuning   dev
±
 docker build -t goodwon593/unsloth-runpod:2025.12.5 .
[+] Building 3.3s (16/16) FINISHED
 => [internal] load build definition from Dockerfile                                                    0.6s
 => => transferring dockerfile: 37B                                                                     0.0s
 => [internal] load .dockerignore                                                                       0.2s
 => => transferring context: 2B                                                                         0.0s
 => [internal] load metadata for docker.io/runpod/pytorch:2.8.0-py3.11-cuda12.8.1-cudnn-devel-ubuntu22.04  2.6s
 => [auth] runpod/pytorch:pull token for registry-1.docker.io                                           0.0s
 => [ 1/10] FROM docker.io/runpod/pytorch:2.8.0-py3.11-cuda12.8.1-cudnn-devel-ubuntu22.04@sha256:cb154fcca15d1d6ce858cfa672  0.0s
 => [internal] load build context                                                                       0.0s
 => => transferring context: 162B                                                                       0.0s
```

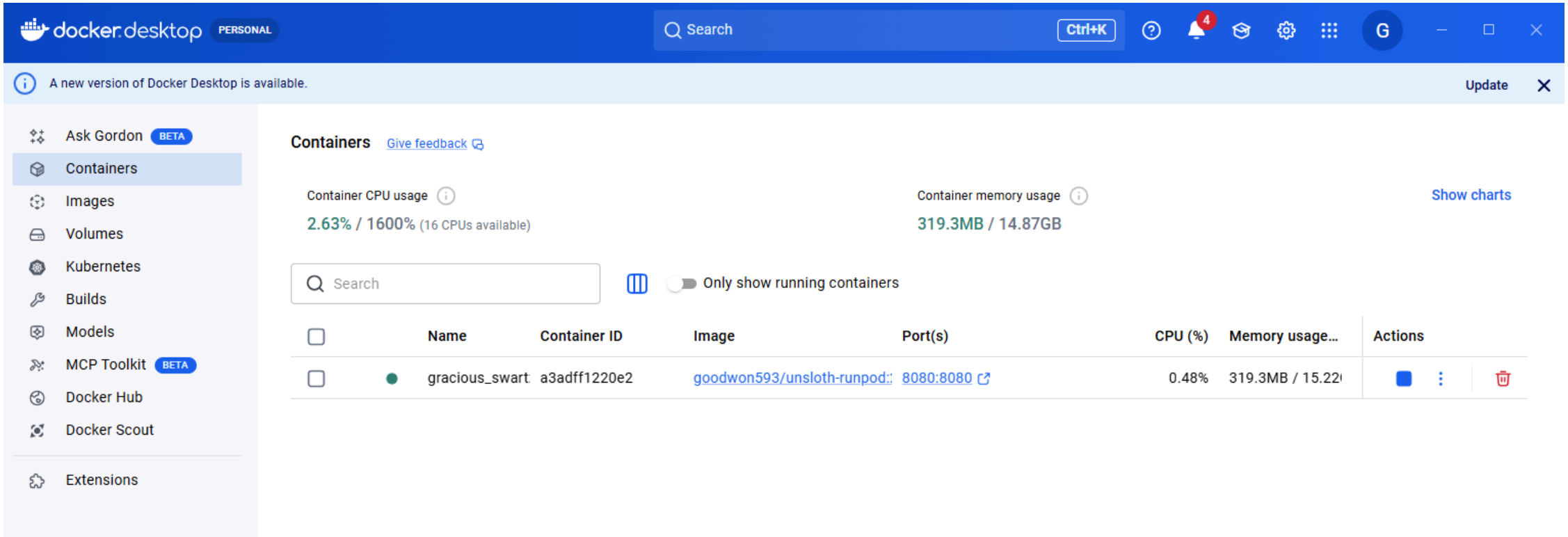
결과 확인


Docker Desktop
Update to latest
Search
⌘K
goodwon593
Containers
Images
Volumes
Dev Environments BETA
Extensions BETA
Add Extensions
Images
Give feedback
An image is a read-only template with instructions for creating a Docker container. Learn more
LOCAL
REMOTE REPOSITORIES
0 Bytes / 7.87 GB in use
1 images
Last refresh: 3 months ago
Search
NAME
TAG
STATUS
CREATED
SIZE
ACTIONS
goodwon593/unsloth-runpod
5c66940e846c
2025.12.5
Unused
3 minutes ago
27.39 GB

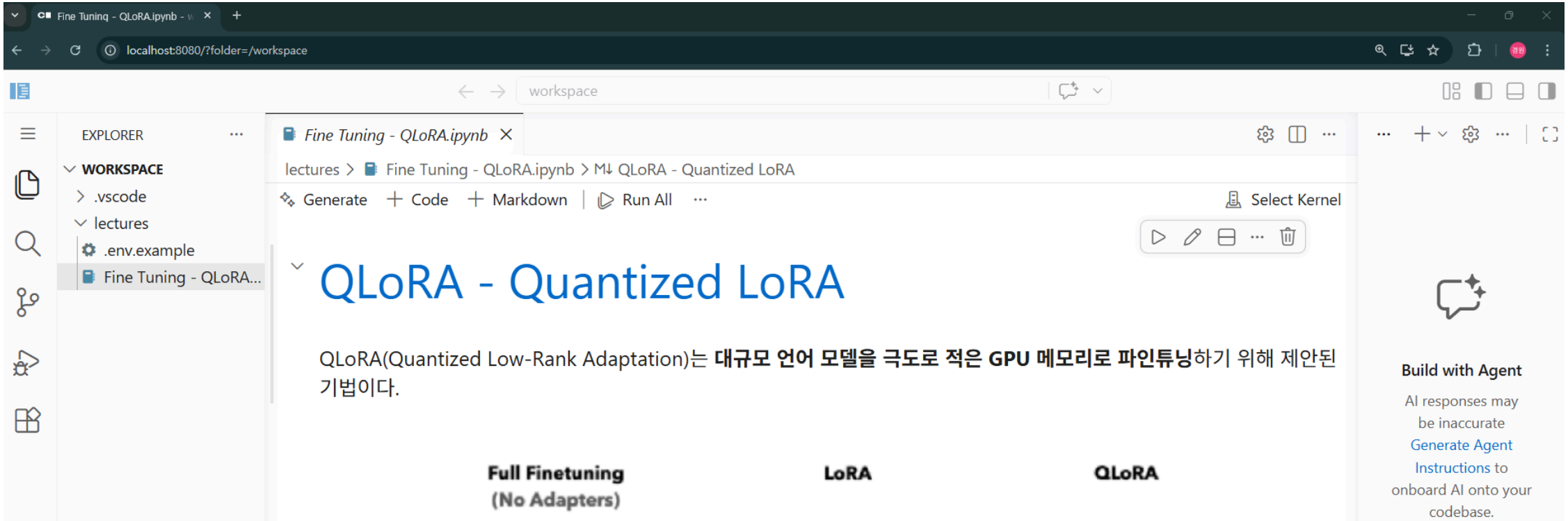
단계2: (옵션)Docker Container 실행

```
# GPU용
docker run -it --rm --gpus all -p 8080:8080 -v ./workspace:/workspace <Docker ID>/unsloth-runpod:2025.12.5
# CPU용
docker run -it --rm -p 8080:8080 -v ./workspace:/workspace <Docker ID>/unsloth-runpod:2025.12.5
```

```
Problems   Output   Debug Console   Terminal   Ports                                   docker

PS C:\dev\course\course_LLM\2. LLM Fine Tuning\2. Fine Tuning with Runpod\3. 모델 학습\1. Unsloth - Fine Tuning> docker run -it --rm -p 8080:8080 -v ./work
space:/workspace goodwon593/unsloth-runpod:2025.12.5

==========
== CUDA ==
==========

CUDA Version 12.8.1

Container image Copyright (c) 2016-2023, NVIDIA CORPORATION & AFFILIATES. All rights reserved.

This container image and its contents are governed by the NVIDIA Deep Learning Container License.
By pulling and using the container, you accept the terms and conditions of this license:
https://developer.nvidia.com/ngc/nvidia-deep-learning-container-license
```


docker desktop
PERSONAL
Search
Ctrl+K
4
G
A new version of Docker Desktop is available.
Update
Ask Gordon
BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Docker Scout
Extensions
Containers
Give feedback
Container CPU usage
2.63% / 1600% (16 CPUs available)
Container memory usage
319.3MB / 14.87GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Memory usage...
Actions
gracious_swart
a3adff1220e2
goodwon593/unsloth-runpod:
8080:8080
0.48%
319.3MB / 15.22

결과 확인

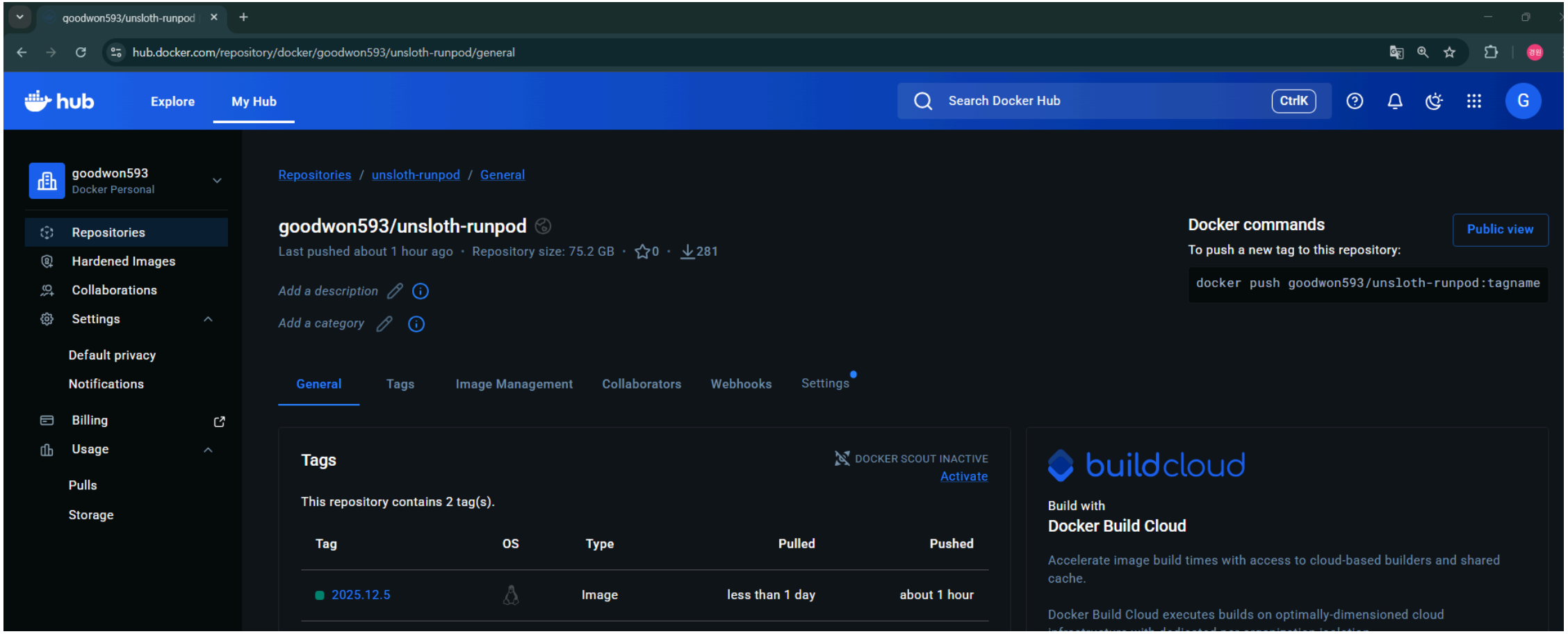

Fine Tuning - QLoRA.ipynb
localhost:8080/?folder=/workspace
workspace
EXPLORER
WORKSPACE
.vscode
lectures
.env.example
Fine Tuning - QLoRA...
Fine Tuning - QLoRA.ipynb
lectures > Fine Tuning - QLoRA.ipynb > M↓ QLoRA - Quantized LoRA
Generate
Code
Markdown
Run All
Select Kernel
QLoRA - Quantized LoRA
QLoRA(Quantized Low-Rank Adaptation)는 대규모 언어 모델을 극도로 적은 GPU 메모리로 파인튜닝하기 위해 제안된 기법이다.
Full Finetuning
(No Adapters)
LoRA
QLoRA
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.

단계3: Docker Hub 배포

```
docker push <Docker ID>/unsloth-runpod:2025.12.5
```

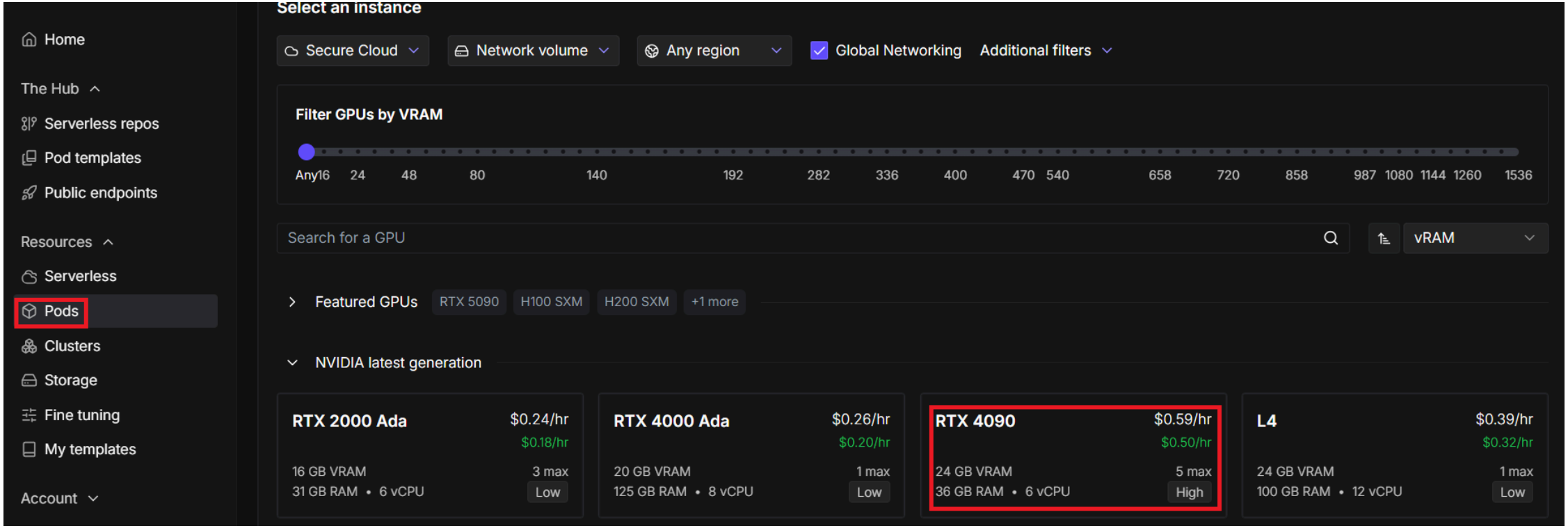
```
Problems   Output   Debug Console   Terminal   Ports   Postman Console                    docker
good593 🚀   ~/dev/github/course_LLM/2. LLM Fine Tuning/2. Fine Tuning with Runpod/3. 모델 학습/1. Unsloth - Fine Tuning   dev ±
docker push goodwon593/unsloth-runpod:2025.12.5
The push refers to repository [docker.io/goodwon593/unsloth-runpod]
6b9c16c2f2bd: Preparing
3bfa851ff8de: Preparing
e5594a072375: Preparing
ccba494f36eb: Preparing
923c65704887: Preparing
3347d6391316: Preparing
82286c484396: Preparing
bd6b8cf096fa: Preparing
82286c484396: Pushing [==============================================>    ]  20.02MB/21.62MB
3ed2c3e5203c: Layer already exists
```

결과확인

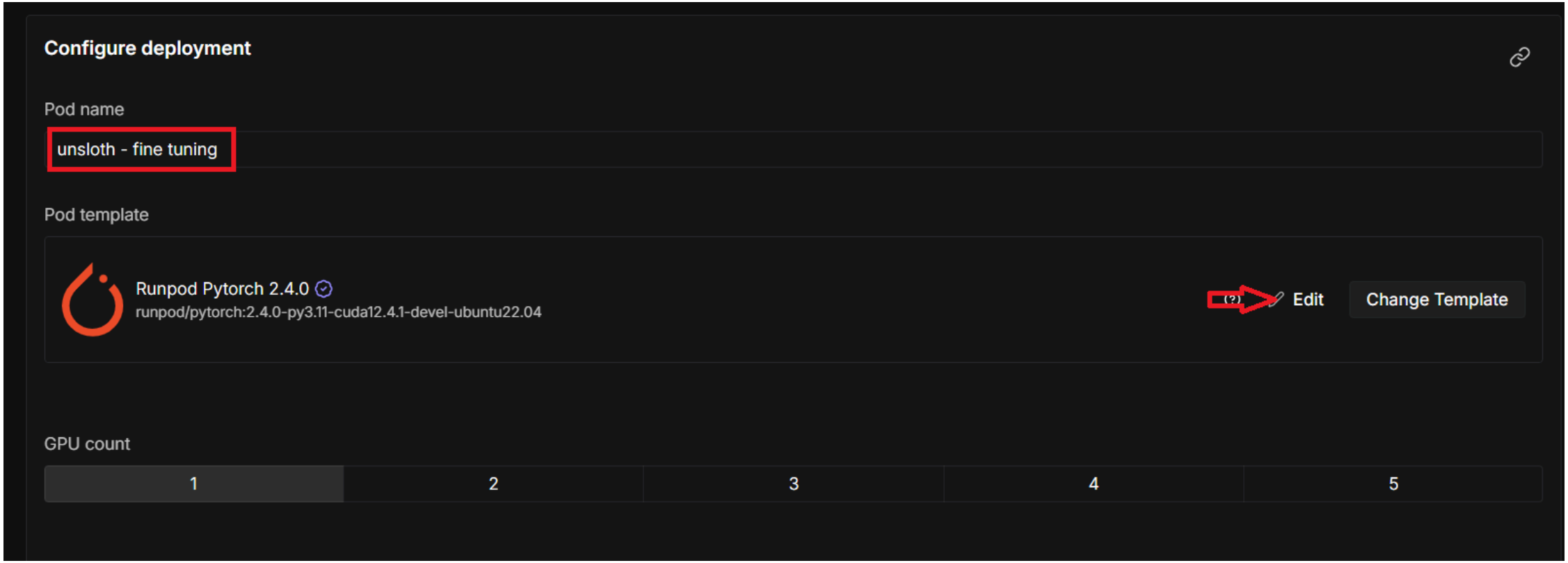

goodwon593/unsloth-runpod
hub.docker.com/repository/docker/goodwon593/unsloth-runpod/general
hub
Explore
My Hub
Search Docker Hub
CtrlK
goodwon593
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Repositories / unsloth-runpod / General
goodwon593/unsloth-runpod
Last pushed about 1 hour ago · Repository size: 75.2 GB · 0 · 281
Add a description
Add a category
Docker commands
Public view
To push a new tag to this repository:
docker push goodwon593/unsloth-runpod:tagname
General
Tags
Image Management
Collaborators
Webhooks
Settings
Tags
DOCKER SCOUT INACTIVE
Activate
This repository contains 2 tag(s).
Tag
OS
Type
Pulled
Pushed
2025.12.5
Image
less than 1 day
about 1 hour
buildcloud
Build with
Docker Build Cloud
Accelerate image build times with access to cloud-based builders and shared cache.
Docker Build Cloud executes builds on optimally-dimensioned cloud

Runpod

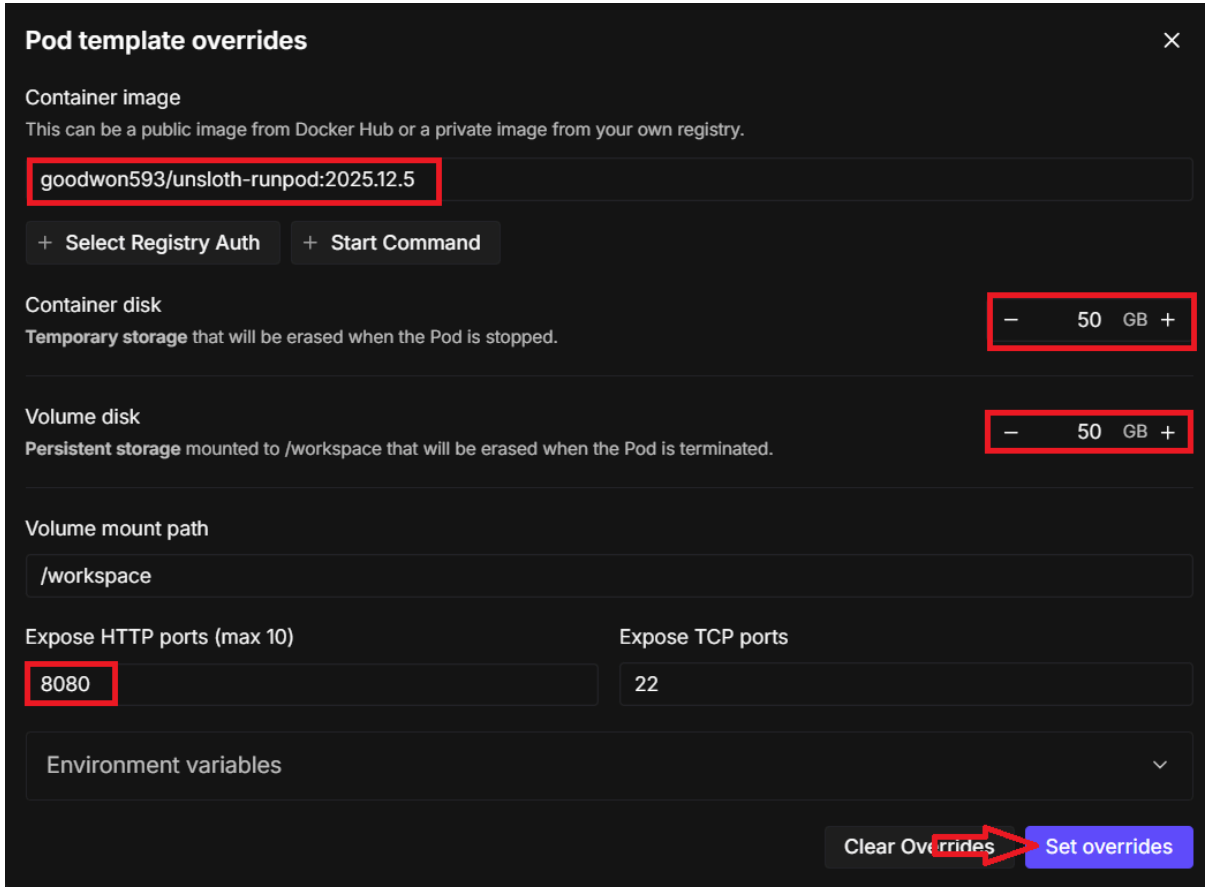
단계1: GPU Pod


Select an instance
Home
The Hub
Serverless repos
Pod templates
Public endpoints
Resources
Serverless
Pods
Clusters
Storage
Fine tuning
My templates
Account
Secure Cloud
Network volume
Any region
Global Networking
Additional filters
Filter GPUs by VRAM
Any16 24 48 80 140 192 282 336 400 470 540 658 720 858 987 1080 1144 1260 1536
Search for a GPU
vRAM
Featured GPUs
RTX 5090
H100 SXM
H200 SXM
+1 more
NVIDIA latest generation
RTX 2000 Ada
$0.24/hr
$0.18/hr
16 GB VRAM
31 GB RAM • 6 vCPU
3 max
Low
RTX 4000 Ada
$0.26/hr
$0.20/hr
20 GB VRAM
125 GB RAM • 8 vCPU
1 max
Low
RTX 4090
$0.59/hr
$0.50/hr
24 GB VRAM
36 GB RAM • 6 vCPU
5 max
High
L4
$0.39/hr
$0.32/hr
24 GB VRAM
100 GB RAM • 12 vCPU
1 max
Low

단계2: Configure deployment



단계3: Pod template overrides



Looking for multi-year savings plans? Contact sales

- [ ] Encrypt volume
- [ ] SSH terminal access
- [ ] Start Jupyter notebook

Storage configuration

Container disk Override

Temporary storage that will be erased when the Pod is stopped.

– 50 GB +

Network volume

$ Save up to 75%

Persistent storage mounted to /workspace that is independent from the Pod's lifecycle.

- Will **not** be deleted when Pod is terminated
- Can be mounted to multiple Pods & Endpoints

$0.07/GB/mo (under 1TB) or $0.05/GB/mo (over 1TB)

Volume disk

Persistent storage mounted to /workspace tied to the Pod's lifecycle.

- **Will be deleted when Pod is terminated**
- Cannot be accessed by other Pods or Endpoints

$0.10/GB/mo when running, $0.20/GB/mo when stopped

Volume disk Override

Persistent storage mounted to /workspace that will be erased when the Pod is terminated.

– 50 GB +

Pricing summary

GPU cost **$0.59 / hr**

Running Pod disk cost **$0.014 / hr**

Stopped Pod disk cost **$0.014 / hr**

Pod summary

1x RTX 4090 (24 GB VRAM)

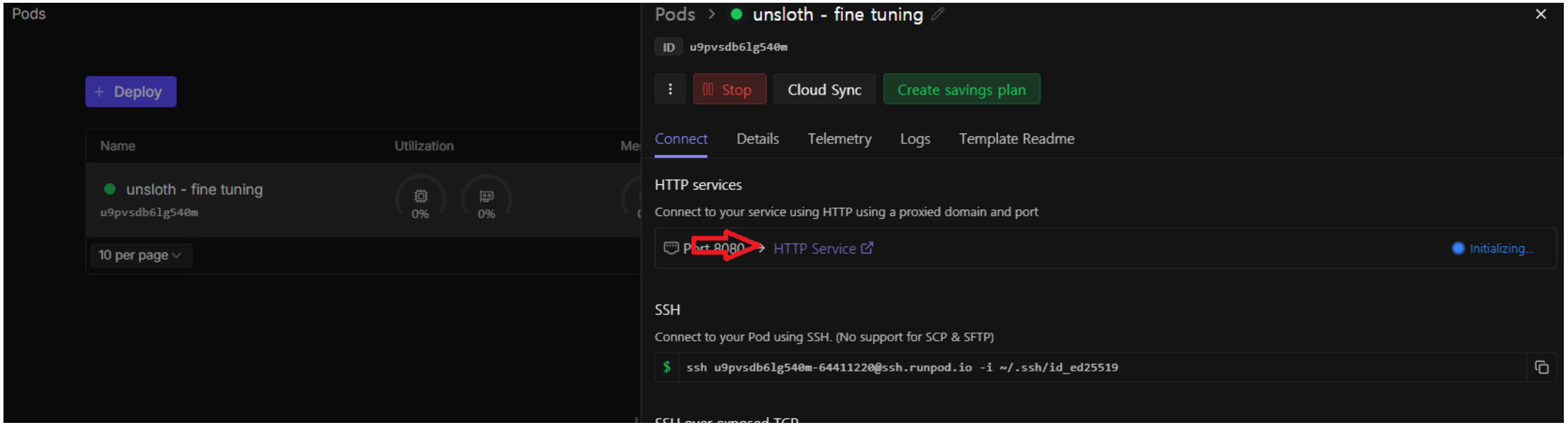
36 GB RAM • 6 vCPU

Total disk 100 GB

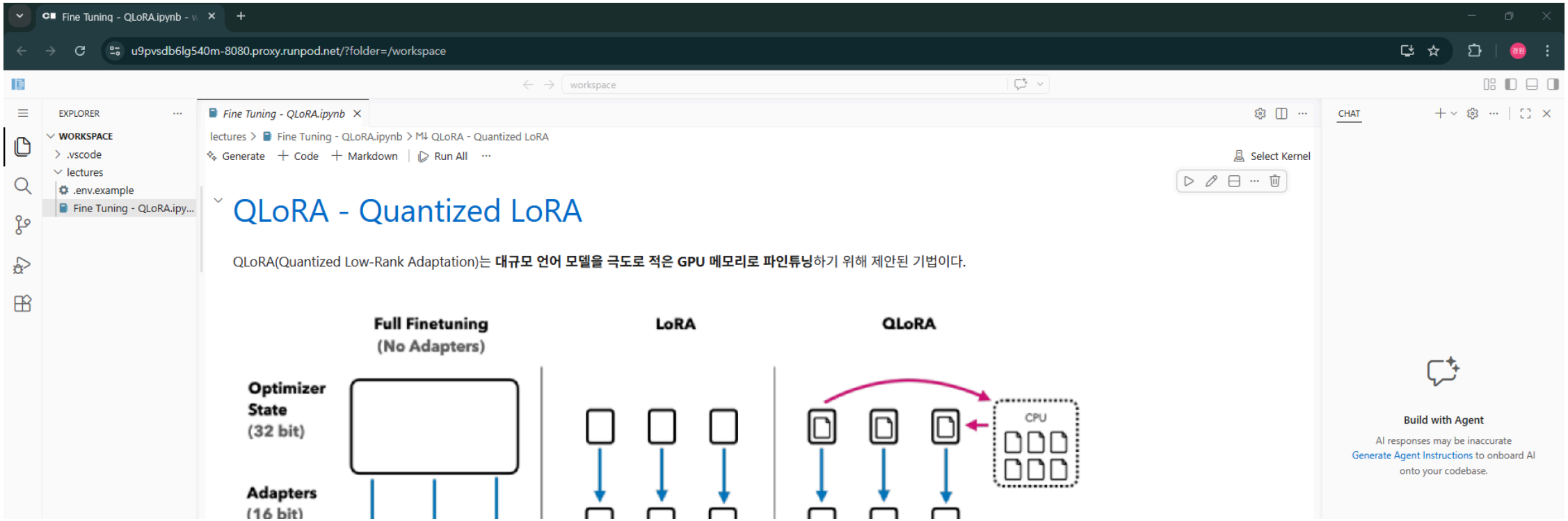
Template overrides applied

Deploy On-Demand

단계4: Jupyter Lab 접속



결과 확인


Fine Tuning - QLoRA.ipynb
u9pvsdb6lg540m-8080.proxy.runpod.net/?folder=/workspace
workspace
EXPLORER
WORKSPACE
.vscode
lectures
.env.example
Fine Tuning - QLoRA.ipy...
Fine Tuning - QLoRA.ipynb
lectures > Fine Tuning - QLoRA.ipynb > QLoRA - Quantized LoRA
Generate
Code
Markdown
Run All
Select Kernel
QLoRA - Quantized LoRA
QLoRA(Quantized Low-Rank Adaptation)는 대규모 언어 모델을 극도로 적은 GPU 메모리로 파인튜닝하기 위해 제안된 기법이다.
Full Finetuning
(No Adapters)
LoRA
QLoRA
Optimizer
State
(32 bit)
Adapters
(16 bit)
CPU
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI
onto your codebase.